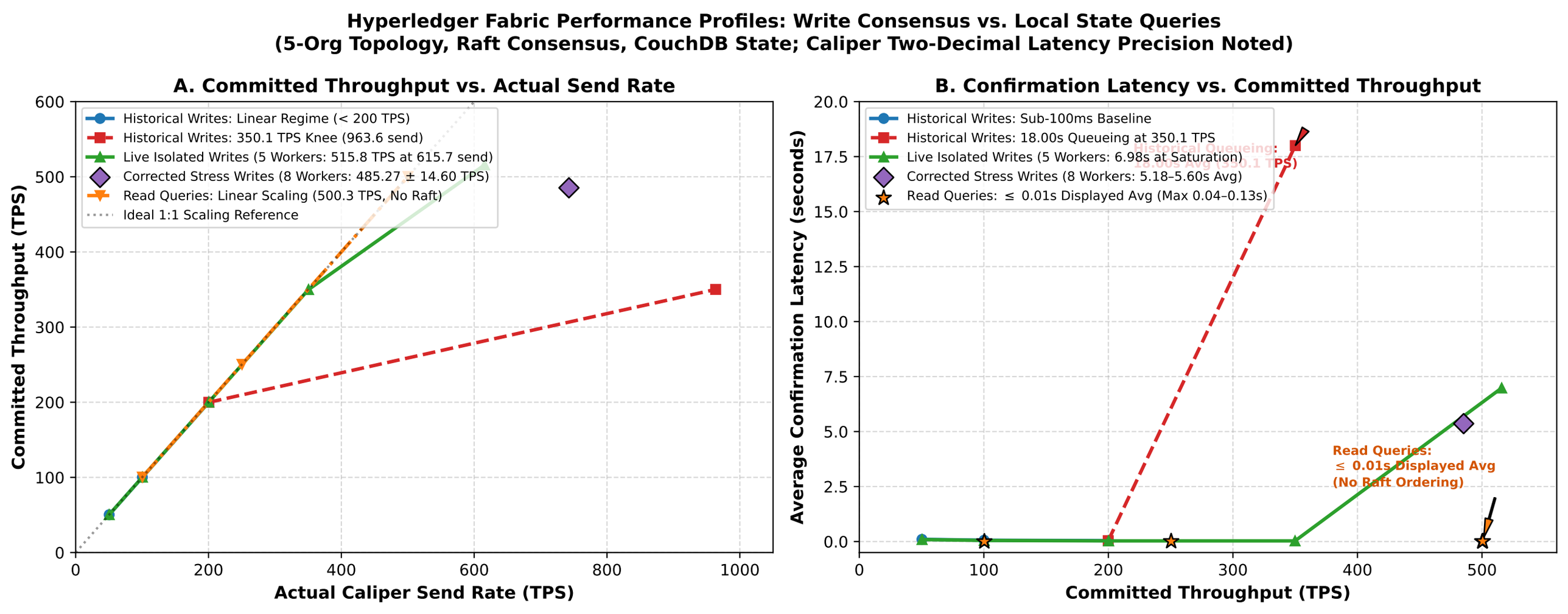
Hyperledger Fabric Performance Profiles: Write Consensus vs. Local State Queries
(5-Org Topology, Raft Consensus, CouchDB State; Caliper Two-Decimal Latency Precision Noted)
A. Committed Throughput vs. Actual Send Rate
Historical Writes: Linear Regime (< 200 TPS)
Historical Writes: 350.1 TPS Knee (963.6 send)
Live Isolated Writes (5 Workers: 515.8 TPS at 615.7 send)
Corrected Stress Writes (8 Workers: 485.27 ± 14.60 TPS)
Read Queries: Linear Scaling (500.3 TPS, No Raft)
Ideal 1:1 Scaling Reference
Committed Throughput (TPS)
Actual Caliper Send Rate (TPS)
0
100
200
300
400
500
600
0
200
400
600
800
1000
B. Confirmation Latency vs. Committed Throughput
Historical Writes: Sub-100ms Baseline
Historical Writes: 18.00s Queueing at 350.1 TPS
Live Isolated Writes (5 Workers: 6.98s at Saturation)
Corrected Stress Writes (8 Workers: 5.18–5.60s Avg)
Read Queries: ≤ 0.01s Displayed Avg (Max 0.04–0.13s)
Historical Queueing:
18.00s Avg (350.1 TPS)
Read Queries:
≤ 0.01s Displayed Avg
(No Raft Ordering)
Average Confirmation Latency (seconds)
Committed Throughput (TPS)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
100
200
300
400
500